HK IMO Training 2020-2021
Problem Set 6
For 28 November 2020

Problem 21. Let p be an odd prime. Find the number of 6-tuples (a, b, c, d, e, f) of integers between 0 and $p-1$ such that

$$a^2 + b^2 + c^2 \equiv d^2 + e^2 + f^2 \pmod{p}.$$

Problem 22. Find all positive integers n for which the numbers in the set $S = \{1, 2, \ldots, n\}$ can be coloured red and blue, with the following condition being satisfied: The set $S \times S \times S$ contains exactly 2007 ordered triples (x, y, z) such that:

(i) The numbers x, y, z are of the same colour and

(ii) The number $x + y + z$ is divisible by n.

Problem 23. Let p be an odd prime. Prove that the $2^{\frac{p-1}{2}}$ numbers $\pm 1 \pm 2 \pm \cdots \pm \dfrac{p-1}{2}$ represent each nonzero residue class mod p the same number of times. What is this number?

Problem 24. Given a prime $p = 8k + 1$ for some integer k, let r be the remainder when $\dbinom{4k}{k}$ is divided by p. Prove that $\sqrt{r}$ is not an integer.